



# US 12,413,694 B2

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|              |                       |         |             |              |              |              |         |         |                           |              |
|--------------|-----------------------|---------|-------------|--------------|--------------|--------------|---------|---------|---------------------------|--------------|
| (56)         | References Cited      |         |             |              | 2011/0305443 | A1 *         | 12/2011 | Sasaki  | H04N 13/183               |              |
|              | U.S. PATENT DOCUMENTS |         |             |              | 2012/0002014 | A1 *         | 1/2012  | Walsh   | 386/E5.028<br>G06T 19/006 |              |
|              | 9,549,167             | B2 *    | 1/2017      | Ushiki       | H04N 13/178  | 2012/0038745 | A1 *    | 2/2012  | Yu                        | H04N 13/183  |
|              | 9,612,657             | B2 *    | 4/2017      | Bertram      | H04N 13/156  |              |         |         |                           | 348/46       |
|              | 9,641,826             | B1 *    | 5/2017      | Carlson      | H04N 13/275  | 2012/0062556 | A1 *    | 3/2012  | Yamamoto                  | H04N 13/366  |
|              | 10,595,001            | B2 *    | 3/2020      | Kim          | H04N 13/122  |              |         |         |                           | 345/419      |
|              | 10,616,567            | B1 *    | 4/2020      | Clemens      | G06T 15/20   | 2012/0086711 | A1 *    | 4/2012  | Lee                       | H04N 13/183  |
|              | 10,656,596            | B2 *    | 5/2020      | Callagy      | G03H 1/0005  |              |         |         |                           | 345/419      |
|              | 10,855,965            | B1 *    | 12/2020     | Jiao         | H04N 13/383  | 2012/0086714 | A1 *    | 4/2012  | Yeon                      | H04N 13/361  |
|              | 10,922,870            | B2 *    | 2/2021      | Vaganov      | G06T 15/02   |              |         |         |                           | 345/419      |
| 2004/0233275 | A1 *                  | 11/2004 | Tomita      | G02B 30/25   |              | 2012/0139906 | A1 *    | 6/2012  | Zhang                     | G06T 19/006  |
|              |                       |         |             | 348/42       |              |              |         |         |                           | 345/419      |
| 2004/0252374 | A1 *                  | 12/2004 | Saishu      | H04N 13/305  |              | 2012/0146894 | A1 *    | 6/2012  | Yang                      | H04N 13/279  |
|              |                       |         |             | 359/462      |              |              |         |         |                           | 345/156      |
| 2005/0001833 | A1 *                  | 1/2005  | Kim         | H04N 13/275  |              | 2012/0169730 | A1 *    | 7/2012  | Inoue                     | H04N 13/128  |
|              |                       |         |             | 348/E13.058  |              |              |         |         |                           | 345/419      |
| 2005/0089212 | A1 *                  | 4/2005  | Mashitani   | H04N 13/366  |              | 2012/0182406 | A1 *    | 7/2012  | Woo                       | H04N 13/398  |
|              |                       |         |             | 348/E13.059  |              |              |         |         |                           | 359/464      |
| 2005/0190258 | A1 *                  | 9/2005  | Siegel      | H04N 13/111  |              | 2012/0229628 | A1 *    | 9/2012  | Ishiyama                  | H04N 13/239  |
|              |                       |         |             | 348/42       |              |              |         |         |                           | 348/135      |
| 2005/0271303 | A1 *                  | 12/2005 | Simpson     | G02B 30/34   |              | 2012/0236002 | A1 *    | 9/2012  | Bi                        | G06T 15/30   |
|              |                       |         |             | 348/E13.067  |              |              |         |         |                           | 345/427      |
| 2007/0188667 | A1 *                  | 8/2007  | Schwerdtner | H04N 13/305  |              | 2012/0249532 | A1 *    | 10/2012 | Kawada                    | H04N 13/144  |
|              |                       |         |             | 348/E13.05   |              |              |         |         |                           | 345/419      |
| 2008/0049100 | A1 *                  | 2/2008  | Lipton      | H04N 13/144  |              | 2012/0256912 | A1 *    | 10/2012 | Yamaguchi                 | H04N 13/128  |
|              |                       |         |             | 348/43       |              |              |         |         |                           | 345/419      |
| 2008/0246757 | A1 *                  | 10/2008 | Ito         | G06T 15/10   |              | 2012/0274629 | A1 *    | 11/2012 | Baek                      | H04N 13/128  |
|              |                       |         |             | 348/E13.008  |              |              |         |         |                           | 345/419      |
| 2009/0219283 | A1 *                  | 9/2009  | Hendrickson | H04N 13/128  |              | 2012/0306860 | A1 *    | 12/2012 | Hatta                     | H04N 13/366  |
|              |                       |         |             | 345/473      |              |              |         |         |                           | 345/419      |
| 2009/0295790 | A1 *                  | 12/2009 | Pockett     | H04N 13/239  |              | 2012/0307023 | A1 *    | 12/2012 | Freiburg                  | H04N 13/128  |
|              |                       |         |             | 345/419      |              |              |         |         |                           | 348/51       |
| 2010/0039502 | A1 *                  | 2/2010  | Robinson    | H04N 13/246  |              | 2012/0320047 | A1 *    | 12/2012 | Yanagita                  | G06F 3/0418  |
|              |                       |         |             | 348/47       |              |              |         |         |                           | 345/419      |
| 2010/0142801 | A1 *                  | 6/2010  | Koppal      | H04N 13/10   |              | 2013/0010059 | A1 *    | 1/2013  | Kobayashi                 | G06T 19/20   |
|              |                       |         |             | 382/154      |              |              |         |         |                           | 348/E13.064  |
| 2010/0142924 | A1 *                  | 6/2010  | Yamashita   | G11B 27/10   |              | 2013/0021458 | A1 *    | 1/2013  | Inoue                     | G02B 30/24   |
|              |                       |         |             | 348/42       |              |              |         |         |                           | 348/56       |
| 2010/0220178 | A1 *                  | 9/2010  | Takahashi   | H04N 13/10   |              | 2013/0038606 | A1 *    | 2/2013  | Ushiki                    | H04N 13/351  |
|              |                       |         |             | 348/54       |              |              |         |         |                           | 348/46       |
| 2010/0253680 | A1 *                  | 10/2010 | Kobayashi   | H04N 13/189  |              | 2013/0050196 | A1 *    | 2/2013  | Takashima                 | H04N 13/376  |
|              |                       |         |             | 345/419      |              |              |         |         |                           | 345/419      |
| 2010/0272417 | A1 *                  | 10/2010 | Nagasawa    | H04N 13/183  |              | 2013/0050197 | A1 *    | 2/2013  | Oosawa                    | H04N 13/327  |
|              |                       |         |             | 386/239      |              |              |         |         |                           | 345/419      |
| 2010/0309296 | A1 *                  | 12/2010 | Harrold     | H04N 13/317  |              | 2013/0063576 | A1 *    | 3/2013  | Okubo                     | G02B 30/33   |
|              |                       |         |             | 348/54       |              |              |         |         |                           | 348/51       |
| 2010/0328533 | A1 *                  | 12/2010 | Mitani      | H04N 13/356  |              | 2013/0070052 | A1 *    | 3/2013  | Yamashita                 | H04N 13/178  |
|              |                       |         |             | 348/521      |              |              |         |         |                           | 348/43       |
| 2011/0012995 | A1 *                  | 1/2011  | Watanabe    | H04N 13/239  |              | 2013/0107014 | A1 *    | 5/2013  | Masuda                    | H04N 13/239  |
|              |                       |         |             | 348/51       |              |              |         |         |                           | 348/47       |
| 2011/0050864 | A1 *                  | 3/2011  | Bond        | H04N 13/261  |              | 2013/0148944 | A1 *    | 6/2013  | Ando                      | G11B 27/034  |
|              |                       |         |             | 348/42       |              |              |         |         |                           | 386/278      |
| 2011/0063421 | A1 *                  | 3/2011  | Kubota      | H04N 13/10   |              | 2013/0162641 | A1 *    | 6/2013  | Zhang                     | H04N 13/128  |
|              |                       |         |             | 348/E13.001  |              |              |         |         |                           | 345/419      |
| 2011/0090215 | A1 *                  | 4/2011  | Ohta        | H04N 13/275  |              | 2013/0165186 | A1 *    | 6/2013  | Choi                      | H04N 13/128  |
|              |                       |         |             | 345/419      |              |              |         |         |                           | 455/566      |
| 2011/0157313 | A1 *                  | 6/2011  | Chiang      | H04N 21/4782 |              | 2013/0169753 | A1 *    | 7/2013  | Lee                       | H04N 13/178  |
|              |                       |         |             | 348/51       |              |              |         |         |                           | 348/43       |
| 2011/0161843 | A1 *                  | 6/2011  | Bennett     | H04N 21/235  |              | 2013/0170737 | A1 *    | 7/2013  | Arita                     | H04N 13/239  |
|              |                       |         |             | 715/760      |              |              |         |         |                           | 382/154      |
| 2011/0175907 | A1 *                  | 7/2011  | Tokuda      | H04N 13/128  |              | 2013/0187852 | A1 *    | 7/2013  | Ebina                     | H04N 13/398  |
|              |                       |         |             | 345/419      |              |              |         |         |                           | 345/158      |
| 2011/0181593 | A1 *                  | 7/2011  | Hirai       | H04N 13/128  |              | 2013/0194253 | A1 *    | 8/2013  | Ohmi                      | H04N 13/128  |
|              |                       |         |             | 382/190      |              |              |         |         |                           | 345/419      |
| 2011/0211815 | A1 *                  | 9/2011  | Yamashita   | H04N 13/128  |              | 2013/0215106 | A1 *    | 8/2013  | Yamaguchi                 | H04N 13/128  |
|              |                       |         |             | 386/E5.028   |              |              |         |         |                           | 345/419      |
| 2011/0229012 | A1 *                  | 9/2011  | Singhal     | H04N 13/111  |              | 2013/0215237 | A1 *    | 8/2013  | Inoue                     | H04N 13/204  |
|              |                       |         |             | 382/154      |              |              |         |         |                           | 345/419      |
| 2011/0242286 | A1 *                  | 10/2011 | Pace        | H04N 13/133  |              | 2013/0229407 | A1 *    | 9/2013  | Chiu                      | H04N 13/31   |
|              |                       |         |             | 348/47       |              |              |         |         |                           | 345/419      |
| 2011/0242296 | A1 *                  | 10/2011 | Ishihara    | H04N 13/183  |              | 2013/0242068 | A1 *    | 9/2013  | Horiuchi                  | G02B 27/0093 |
|              |                       |         |             | 348/54       |              |              |         |         |                           | 348/55       |
| 2011/0292183 | A1 *                  | 12/2011 | Tajiri      | H04N 13/189  |              | 2013/0268882 | A1 *    | 10/2013 | Roh                       | G06F 3/04815 |
|              |                       |         |             | 348/E13.074  |              |              |         |         |                           | 715/782      |

(56)

**References Cited****U.S. PATENT DOCUMENTS**

2013/0293446 A1 \* 11/2013 Yoo ..... H04N 13/356  
345/6  
2014/0078273 A1 \* 3/2014 Ono ..... G03B 35/26  
348/54  
2014/0085442 A1 \* 3/2014 Ono ..... H04N 13/337  
348/54  
2014/0143733 A1 \* 5/2014 Jung ..... H04N 13/261  
715/848  
2014/0146143 A1 \* 5/2014 Park ..... H04N 13/351  
348/51  
2014/0300711 A1 \* 10/2014 Kroon ..... H04N 13/307  
348/51  
2014/0354796 A1 \* 12/2014 Hein ..... G02B 21/086  
359/385  
2015/0036211 A1 \* 2/2015 Chen ..... G02B 30/31  
359/462  
2015/0334369 A1 \* 11/2015 Bruls ..... H04N 13/128  
348/43  
2016/0070112 A1 \* 3/2016 Song ..... G02B 30/40  
359/471  
2016/0077349 A1 \* 3/2016 An ..... H04N 13/351  
359/462  
2016/0150209 A1 \* 5/2016 Dadash Pour ..... H04N 13/128  
348/51  
2016/0191904 A1 \* 6/2016 An ..... G02B 30/27  
348/51  
2016/0219258 A1 \* 7/2016 Woodgate ..... H04N 23/661  
2016/0234482 A1 \* 8/2016 Bickerstaff ..... H04N 13/128

2016/0241842 A1 \* 8/2016 Newbery ..... G03B 35/10  
2016/0360187 A1 \* 12/2016 Smithwick ..... G02B 30/27  
2017/0038598 A1 \* 2/2017 Song ..... G02B 30/36  
2017/0041596 A1 \* 2/2017 Park ..... H04N 13/305  
2017/0078652 A1 \* 3/2017 Hua ..... G06V 20/20  
2017/0289518 A1 \* 10/2017 Kim ..... H04N 13/239  
2017/0366797 A1 \* 12/2017 Kim ..... H04N 13/128  
2018/0007349 A1 \* 1/2018 Lee ..... G02B 30/27  
2019/0180514 A1 \* 6/2019 Sasaki ..... H04N 13/128  
2020/0029067 A1 \* 1/2020 Lee ..... H04N 13/383  
2020/0126494 A1 \* 4/2020 Chen ..... G09G 5/10  
2020/0201036 A1 \* 6/2020 Hong ..... B60R 1/31  
2021/0132693 A1 \* 5/2021 Pulli ..... G02B 27/0093  
2021/0136354 A1 \* 5/2021 Valli ..... G02B 27/017  
2021/0314542 A1 \* 10/2021 Jung ..... H04N 13/122  
2022/0087813 A1 \* 3/2022 Burge ..... G02C 7/104  
2022/0264076 A1 \* 8/2022 Makinen ..... G02B 30/26  
2022/0277522 A1 \* 9/2022 Katsuki ..... H04N 17/00  
2022/0385880 A1 \* 12/2022 Nims ..... H04N 13/271  
2024/0118550 A1 \* 4/2024 Lai ..... G02B 27/0172  
2024/0169673 A1 \* 5/2024 Xiong ..... G06T 17/205  
2024/0187561 A1 \* 6/2024 Tomizawa ..... H04N 13/32  
2024/0221276 A1 \* 7/2024 Foda ..... H04N 13/128  
2024/0414309 A1 \* 12/2024 He ..... H04N 13/279

**FOREIGN PATENT DOCUMENTS**

|    |           |        |
|----|-----------|--------|
| CN | 114387324 | 4/2022 |
| TW | 201518847 | 5/2015 |
| TW | 201815162 | 4/2018 |
| TW | 202201348 | 1/2022 |

\* cited by examiner

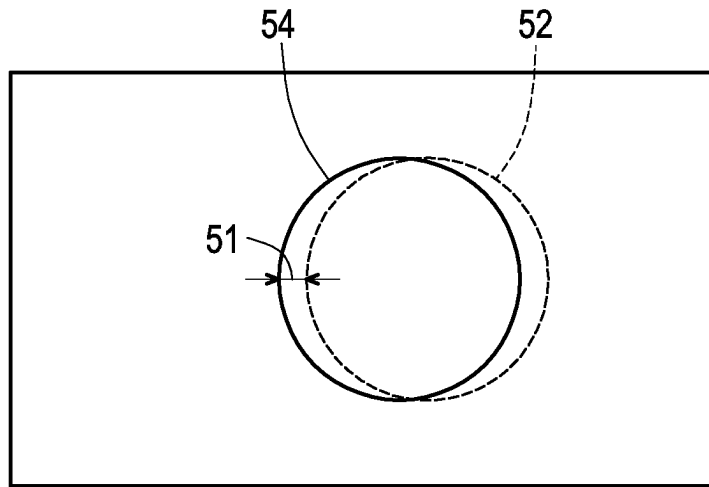


FIG. 1A

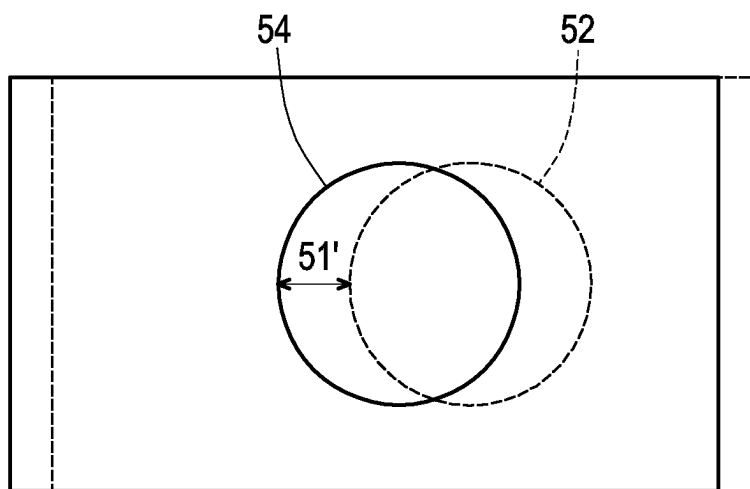


FIG. 1B

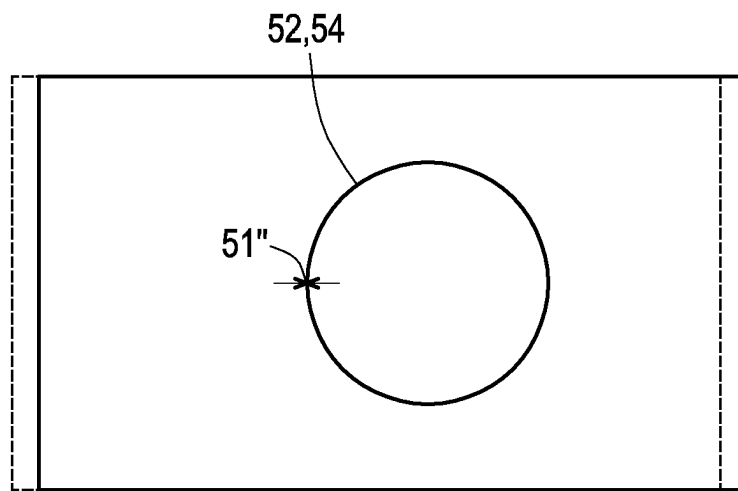


FIG. 1C

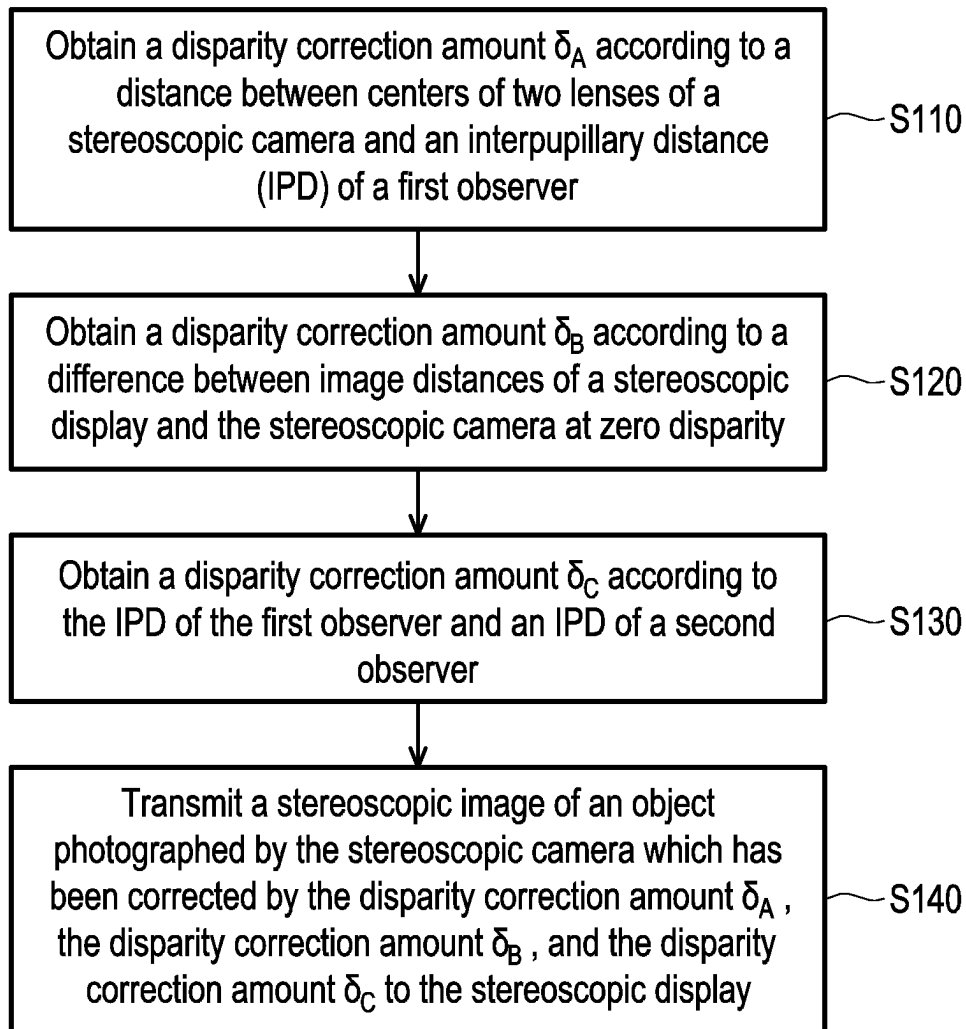


FIG. 2

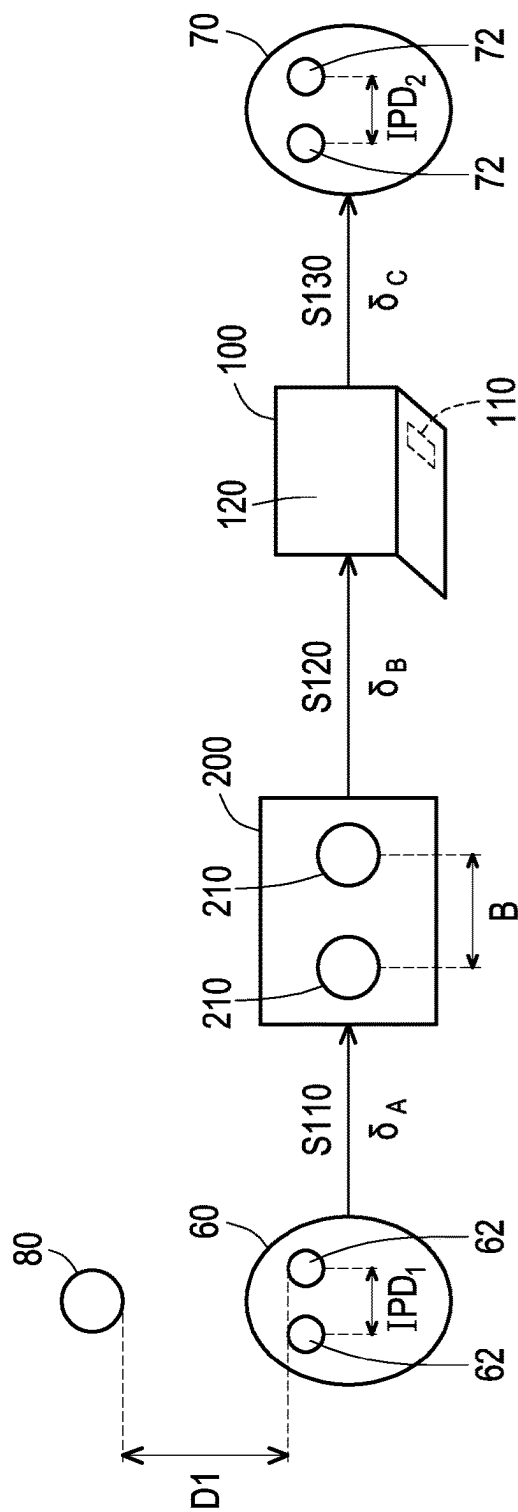


FIG. 3A

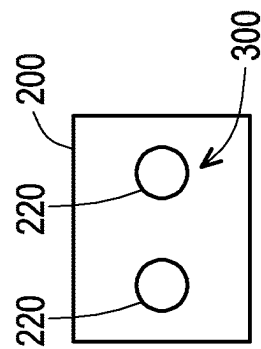


FIG. 3B

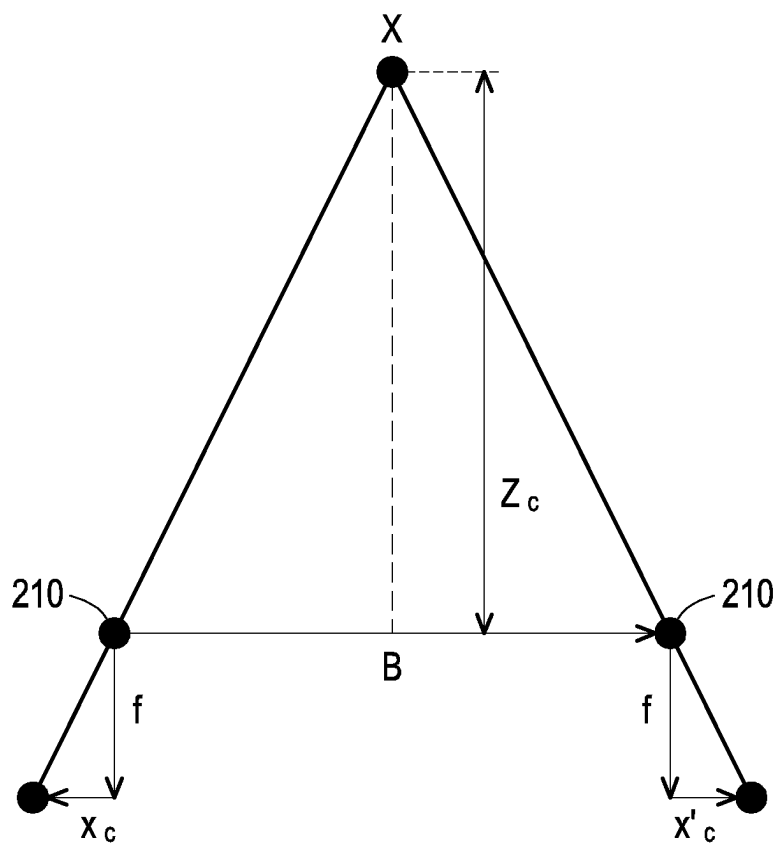


FIG. 4



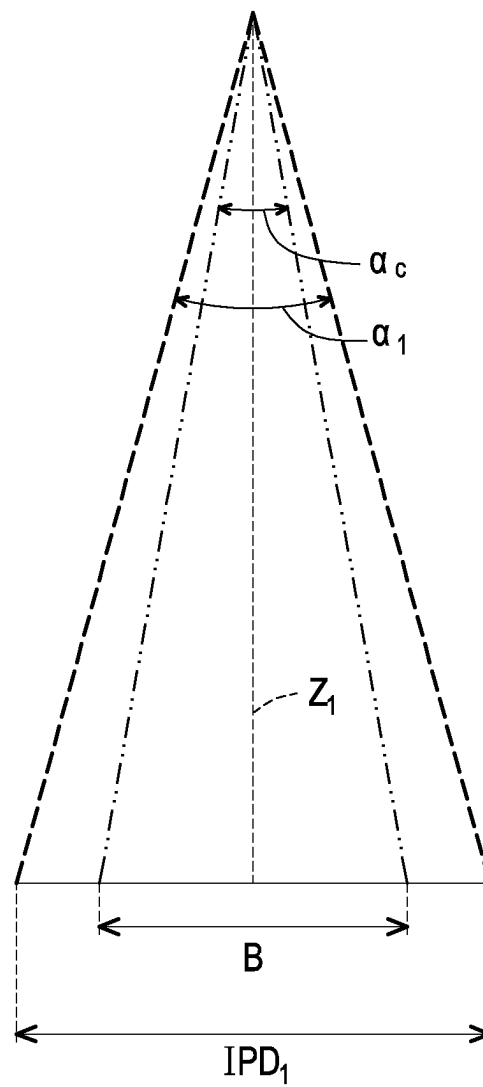


FIG. 5

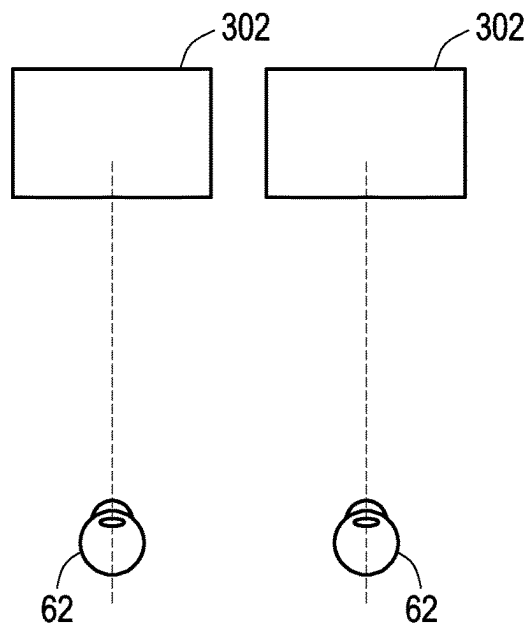


FIG. 6

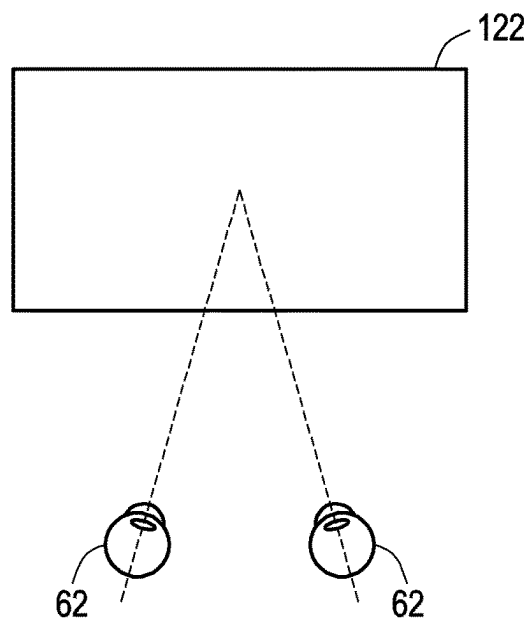


FIG. 7

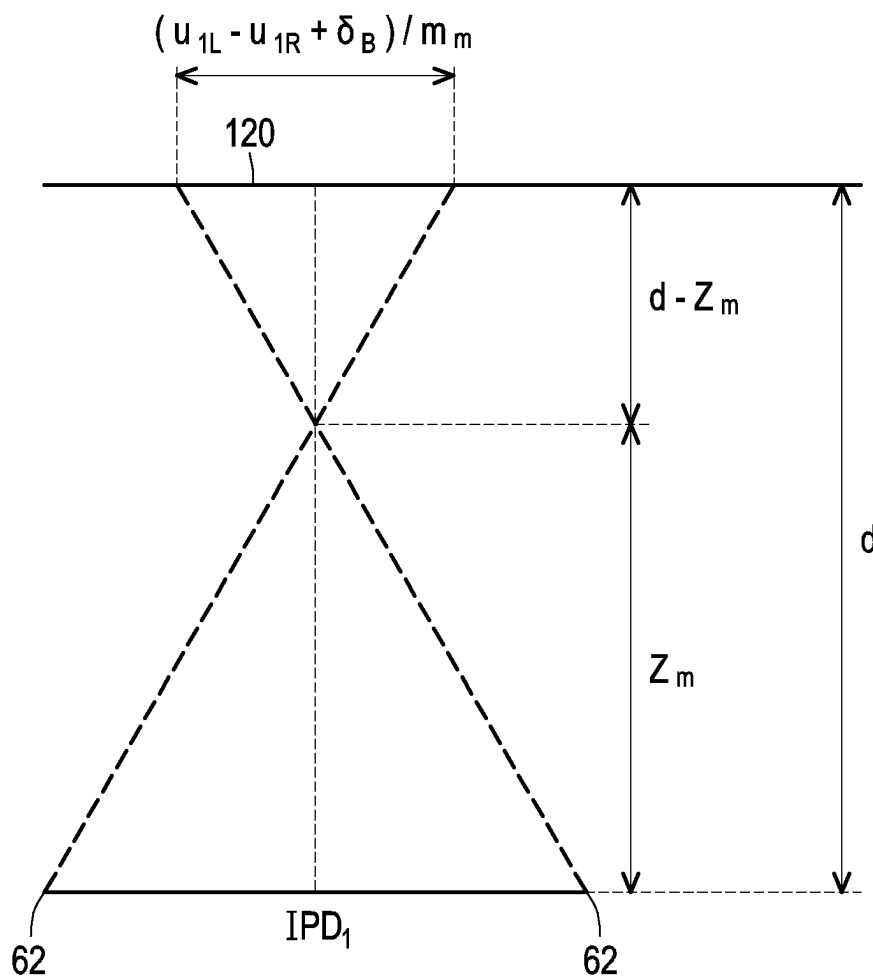


FIG. 8

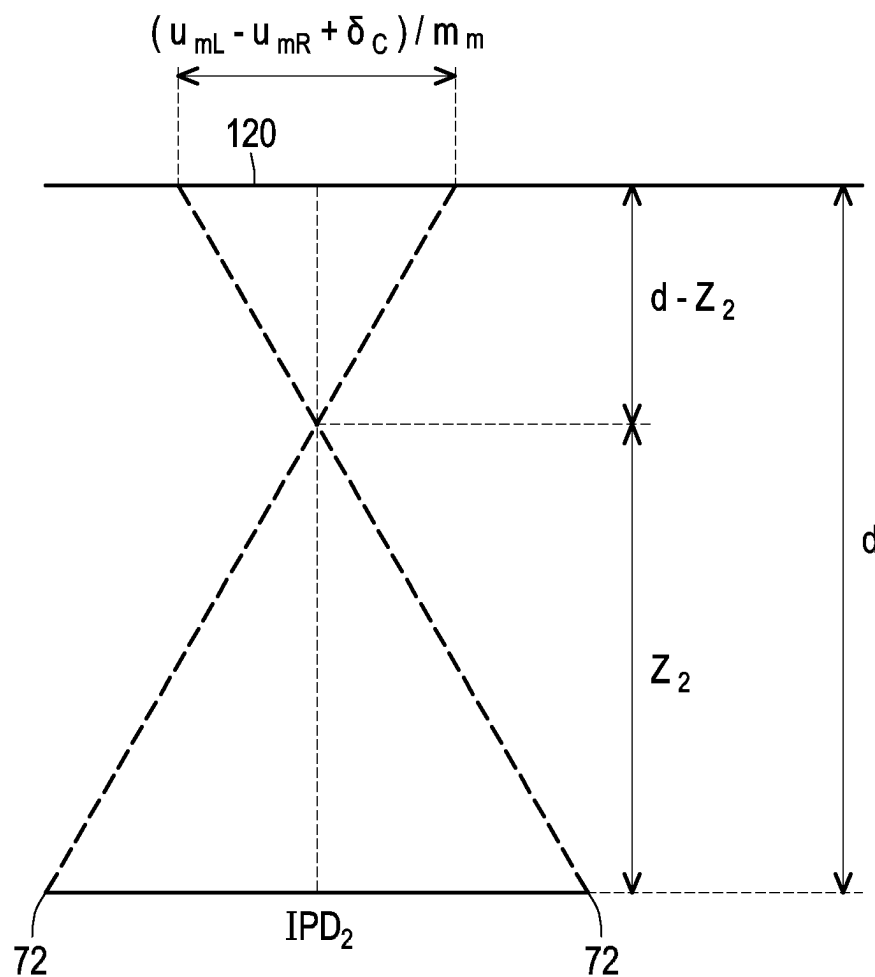


FIG.9

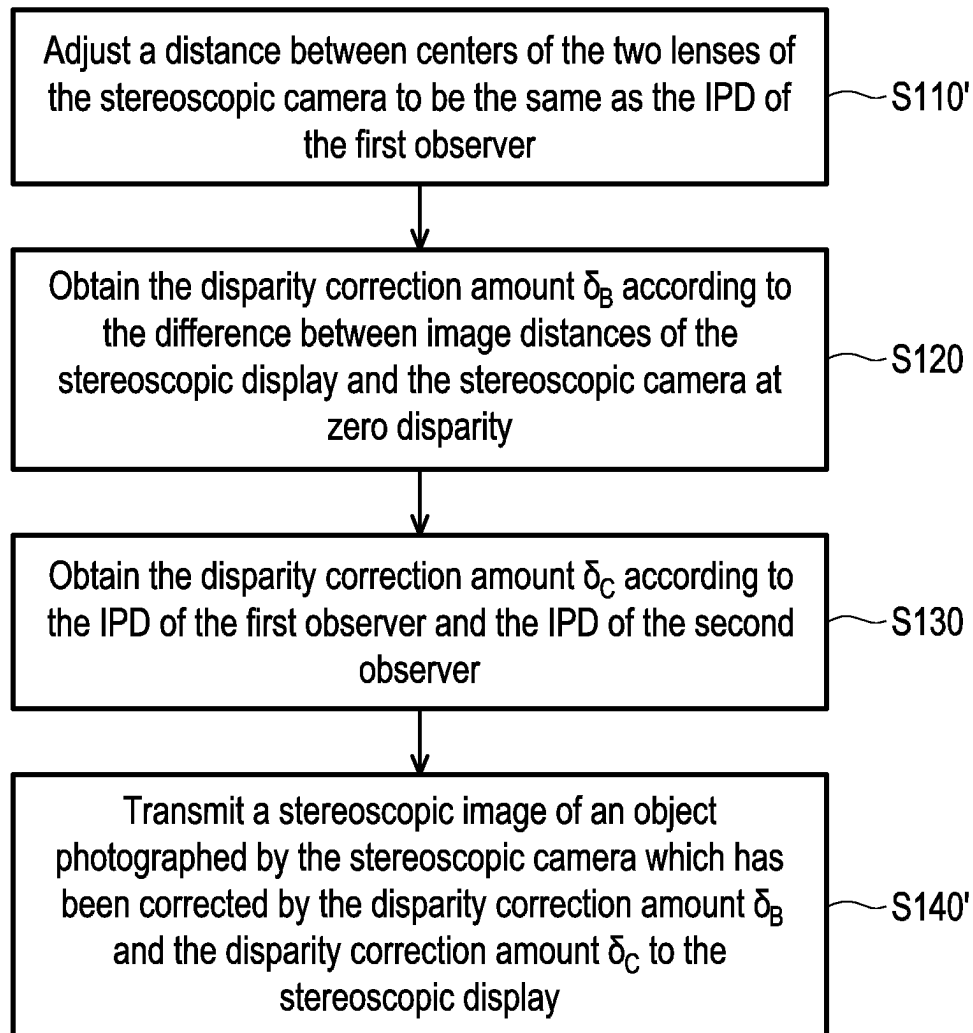


FIG. 10

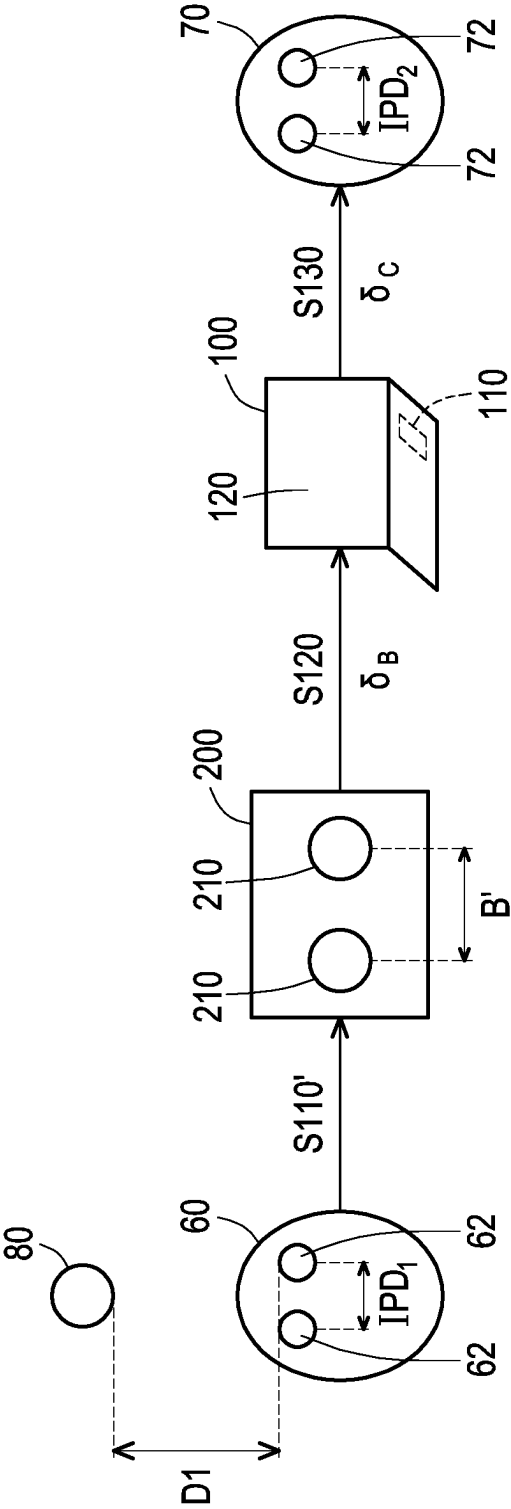


FIG. 11

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## METHOD AND DEVICE FOR ADJUSTING DEPTH OF STEREOSCOPIC IMAGE

### CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of Taiwan application serial no. 111150471, filed on Dec. 28, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

### BACKGROUND

#### Technical Field

The disclosure relates to a method and device for adjusting an image, and particularly relates to a method and device for adjusting a depth of a stereoscopic image.

#### Description of Related Art

The interpupillary distance (IPD) of each person is quite different. For people with a large IPD, a stereoscopic feeling when seeing ordinary objects is more prominent compared with people with a small IPD. For this reason, when using a stereoscopic display product, such as a virtual reality display (VR display), an augmented reality display (AR display), a stereoscopic TV, inputting the IPD of a seer or a user into a software is required to achieve a realistic stereoscopic size and feeling. As in the stereoscopic TV field, image sources photographed by different users have different depth records. When another user sees through a stereoscopic display, since two users have different IPDs, perceived depths are also different.

### SUMMARY

The disclosure provides a method for adjusting a depth of a stereoscopic image, which enables users with different interpupillary distances (IPDs), or stereoscopic image photographers and stereoscopic display viewers all to see stereoscopic images with the same depth effect.

The disclosure provides a device for adjusting a depth of a stereoscopic image, which enables users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers all to see stereoscopic images with the same depth effect.

An embodiment of the disclosure provides a method for adjusting a depth of a stereoscopic image, which includes the following. A first disparity correction amount is obtained according to a distance between centers of two lenses of a stereoscopic camera and an IPD of a first observer, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object. A second disparity correction amount is obtained according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera. A third disparity correction amount is obtained according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image

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distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object. A stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is transmitted to the stereoscopic display.

An embodiment of the disclosure provides a device for adjusting a depth of a stereoscopic image, which includes at least one controller used to receive a stereoscopic image signal from a stereoscopic camera and to control a display content of a stereoscopic display. The controller is configured to execute the following: obtain a first disparity correction amount according to a distance between centers of two lenses of the stereoscopic camera and an IPD of a first observer, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object; obtain a second disparity correction amount according to a difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera; obtain a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and transmit a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

An embodiment of the disclosure provides a method for adjusting a depth of a stereoscopic image, which includes the following. A distance between centers of two lenses of a stereoscopic camera is adjusted to be the same as an IPD of a first observer, so that a stereoscopic image distance of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object. A first disparity correction amount is obtained according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera. A second disparity correction amount is obtained according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image distance formed after adding the first disparity correction amount and the second disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object. A stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount is transmitted to the stereoscopic display.

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In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the first disparity correction amount is obtained according to the distance between centers of the two lenses of the stereoscopic camera and the IPD of the first observer, the second disparity correction amount is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the third disparity correction amount is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect. In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the distance between centers of the two lenses of the stereoscopic camera is adjusted to be the same as the IPD of the first observer, the first disparity correction amount is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the second disparity correction amount is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect.

#### BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

FIG. 1A, FIG. 1B, and FIG. 1C are schematic diagrams of a disparity on a stereoscopic display.

FIG. 2 is a flowchart of a method for adjusting a depth of a stereoscopic image according to an embodiment of the disclosure.

FIG. 3A is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 2.

FIG. 3B is a schematic diagram of another side of the stereoscopic camera in FIG. 3A.

FIG. 4 and FIG. 5 are graphs used to calculate a disparity correction amount  $\delta_A$  in FIG. 2.

FIG. 6, FIG. 7, and FIG. 8 are graphs used to calculate a disparity correction amount  $\delta_B$  in FIG. 2.

FIG. 9 is a graph for calculating a disparity correction amount  $\delta_C$  in FIG. 2.

FIG. 10 is a flowchart of a method for adjusting a depth of a stereoscopic image according to another embodiment of the disclosure.

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FIG. 11 is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image shown in FIG. 10.

#### DESCRIPTION OF THE EMBODIMENTS

FIG. 1A, FIG. 1B, and FIG. 1C are schematic diagrams of a disparity on a stereoscopic display. Please refer to FIG. 1A, FIG. 1B and FIG. 1C. A depth of a stereoscopic image is mainly displayed by using a difference amount of a disparity **51** of two eyes, and the depth mentioned here is a distance between the stereoscopic image and the eyes felt by an observer. An image **52** and an image **54** are images projected to the left eye and the right eye, respectively. When the disparity **51** is large, a stereoscopic feeling generated is strong, which means that the stereoscopic image stands out (that is, the depth is small); on the contrary, when the disparity **51** is small, a stereo vision perception is minor, which means that the depth is large and the stereoscopic image is far away from a user.

If adjusting the depth of the stereoscopic image is desired, the difference amount of a disparity of two eyes may be used to increase a disparity between the image **52** and the image **54** (such as adjusting from the disparity **51** in FIG. 1A to a larger disparity **51'** in FIG. 1B) so that the stereoscopic image may be made to stand out and be close to the user. On the other hand, as in FIG. 1C, if a disparity **51"** is adjusted to 0, the depth of the stereoscopic image is on a display surface of the stereoscopic display.

Therefore, by adjusting the amount of the disparity, the depth of the stereoscopic image may be adjusted.

FIG. 2 is a flowchart of a method for adjusting a depth of a stereoscopic image according to an embodiment of the disclosure. FIG. 3A is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 2. FIG. 3B is a schematic diagram of another side of a stereoscopic camera in FIG. 3A. Please refer to FIG. 2 and FIG. 3A first. This embodiment provides a device for adjusting a depth of a stereoscopic image **100**, which includes at least one controller **110** (one of the at least one controller **110** is taken as an example in FIG. 3A) used to receive a stereoscopic image signal from a stereoscopic camera **200** and to control a display content of a stereoscopic display **120**. The controller **110** is configured to execute Step **S110** to Step **S140** in FIG. 2. In other words, the method for adjusting a depth of a stereoscopic image of this embodiment includes Step **S110** to Step **S140**, and it may be implemented by the device for adjusting a depth of the stereoscopic image **100** of this embodiment.

In the method for adjusting a depth of a stereoscopic image of this embodiment, Step **S110** is first executed as follows. A disparity correction amount  $\delta_A$  is obtained according to a distance **B** between centers of two lenses **210** of the stereoscopic camera **200** and an interpupillary distance (IPD)  $IPD_1$  of a first observer **60**, in which a stereoscopic image distance (that is, a distance between a stereoscopic image and eyes perceived by a user when seeing the stereoscopic image, that is, a depth of the stereoscopic image) formed after adding the disparity correction amount  $\delta_A$  to a disparity of a stereoscopic image of an object **80** photographed by the stereoscopic camera **200** is equal to a distance **D1** (that is, an actual distance between eyes **62** of the first observer **60** and the object **80**) of the first observer **60** seeing the object **80**.

Next, Step **S120** is executed as follows. A disparity correction amount  $\delta_B$  is obtained according to a difference between image distances of the stereoscopic display **120** and



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the stereoscopic camera **200** at zero disparity, in which a stereoscopic image distance formed after adding the disparity correction amount  $\delta_B$  to a disparity of a stereoscopic image displayed by the stereoscopic display **120** is equal to a stereoscopic image distance provided by the stereoscopic camera **200**.

Then, Step **S130** is executed as follows. A disparity correction amount  $\delta_C$  is obtained according to the IPD  $IPD_1$  of the first observer **60** and an IPD  $IPD_2$  (that is, a distance between pupils of two eyes **72** of a second observer **70**) of the second observer **70**, in which a stereoscopic image distance formed after adding the disparity correction amount  $\delta_A$ , the disparity correction amount  $\delta_B$ , and the disparity correction amount  $\delta_C$  to the disparity of the stereoscopic image displayed by the stereoscopic display **120** is equal to the distance **D1** of the first observer **60** seeing the object **80**.

Afterward, Step **S140** is executed as follows. A stereoscopic image of the object **80** photographed by the stereoscopic camera **200** which has been corrected by the disparity correction amount  $\delta_A$ , the disparity correction amount  $\delta_B$ , and the disparity correction amount  $\delta_C$  is transmitted to the stereoscopic display **120**.

In this embodiment, the stereoscopic camera **200** may be integrated with a virtual reality display **300** to display the stereoscopic image of the object **80** photographed by the stereoscopic camera **200**. For example, the front of the stereoscopic camera **200** is provided with the lens **210**, which is able to photograph the object **80**, and the back of the stereoscopic camera **200** is provided with virtual image displays **220**, and the observer **60** is able to see a stereoscopic image generated by virtual reality by aligning the two eyes **62** to the two virtual image displays **220** respectively. The method for adjusting a depth of a stereoscopic image further includes the following. A stereoscopic image of the object **80** photographed by the stereoscopic camera **200** which has been corrected by the disparity correction amount  $\delta_A$  of the stereoscopic image is displayed by the virtual reality display **300**, so that a stereoscopic image distance of a stereoscopic image seen by the first observer **60** through the virtual reality display **300** is equal to the distance **D1** of the first observer **60** directly seeing the object **80**.

In this embodiment, the disparity correction amount  $\delta_A$ , the disparity correction amount  $\delta_B$ , and the disparity correction amount  $\delta_C$  may be calculated by a pixel difference, and the pixel difference refers to a distance of the number of pixels deviated. In addition, in this embodiment, the method of obtaining the disparity correction amount  $\delta_A$ , the disparity correction amount  $\delta_B$ , and the disparity correction amount  $\delta_C$  is performed through calculation with triangulation. The calculation process of Steps **S110**, **S120**, and **S130** will be described in detail below.

FIG. **4** and FIG. **5** are graphs used to calculate the disparity correction amount  $\delta_A$  of FIG. **2**. Please refer to FIG. **3A**, FIG. **4**, and FIG. **5**. When calculating the disparity correction amount  $\delta_A$ , assuming that the first observer **60** utilizes the stereoscopic camera **200** to take a picture or record a video to generate a stereoscopic image file, but due to the IPD  $IPD_1$  of the first observer **60** and the distance **B** between centers of the two lenses **210** of the stereoscopic camera **200** is different, so the disparity correction amount  $\delta_A$  needs to be added at this time. Also, the following mathematical formula is used to calculate.

$$Z_c = \frac{fB}{(x_c - x'_c)} = \frac{fB}{\frac{1}{m_c}(u_L - u_R)} \quad (\text{Formula 1})$$

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$$m_c = \frac{\text{camera\_pixels}}{\text{mm}} \quad (\text{Formula 2})$$

FIG. **4** is a simple structure diagram of the stereoscopic camera **200**, **B** is the distance between centers of the two lenses **210** of the stereoscopic camera **200**, **X** is a certain object point photographed, **x** and **x'** are actual projection positions on a photosensitive element of a camera, **f** is a focal length of the camera, and  $m_c$  is a pixel density. For example, camera\_pixels in Formula 2 is the number of pixels per millimeter (mm) on the photosensitive element. As in Formula 1, a depth  $Z_c$  is composed of a pixel position difference ( $u_L - u_R$ ) projected by **X** on a photo. That is, the depth may be adjusted by adjusting the pixel difference between the left eye and the right eye.

$$Z_c = \frac{B}{\tan \frac{\alpha_c}{2}} \quad (\text{Formula 3})$$

$$\tan \frac{\alpha_c}{2} = \frac{(x_c - x'_c)}{2f} = \frac{1}{m_c} \frac{(u_L - u_R)}{2f} \quad (\text{Formula 4})$$

$$Z_1 = \frac{IPD_1}{\tan \frac{\alpha_1}{2}} \quad (\text{Formula 5})$$

$$Z_c = Z_1 \quad (\text{Formula 6})$$

$$\alpha_1 = \alpha_c + \varphi_1 \quad (\text{Formula 7})$$

$$\tan \frac{\varphi_1}{2} = \frac{\frac{IPD_1}{B} - 1}{\frac{1}{\tan \frac{\alpha_c}{2}} + \frac{IPD_1}{B} \tan \frac{\alpha_c}{2}} \quad (\text{Formula 8})$$

$$Disparity_1 = (u_L - u_R + \delta_A) \quad (\text{Formula 9})$$

$$\delta_A = PPD_{cm} \cdot \varphi_1 \quad (\text{Formula 10})$$

Assuming that the stereoscopic camera **200** is integrated with the virtual reality display **300**, the first observer **60** sees a content photographed by oneself through the virtual reality display **300**. The IPD  $IPD_1$  of the first observer **60** is different from **B**, as in FIG. **5**. Since a stereoscopic image distance is generated by a disparity, and a virtual image imaging distance of the virtual reality display **300** may be infinite, it is necessary to convert the unit of the disparity from mm into an angle deg. A difference amount of the disparity is  $\varphi_1$ , as in Formula 7. Also, the purpose is that a depth  $Z_1$  seen by the first observer **60** is equal to the depth  $Z_c$  photographed. From Formula 3 and Formula 5, Formula 8 may be obtained to obtain  $\varphi_1$ . In addition,  $\varphi_1$  is converted into a pixel difference (i.e., the disparity correction amount  $\delta_A$ ), as in Formula 10, and the conversion of the angle into the pixel is calculated by the pixel per deg  $PPD_{cm}$  of the virtual reality display **300**. In this way, a difference amount of the pixel difference  $\delta_A$  may be increased. That is, a depth photographed by the stereoscopic camera **200** is the same as a depth seen by the first observer **60**. A new disparity  $Disparity_1$  (i.e., ( $u_L - u_R$ )) is described as in Formula 9.

FIG. **6**, FIG. **7**, and FIG. **8** are graphs used to calculate the disparity correction amount  $\delta_B$  in FIG. **2**. Please refer to FIG. **3A**, FIG. **6**, FIG. **7**, and FIG. **8**. When calculating the disparity correction amount  $\delta_A$ , as illustrated in FIG. **6**, since placements of the virtual reality display **300** and the stereo-

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stereoscopic camera **200** are parallel placements, under such a structure, when the disparity is 0, the equivalent is a two-dimensional plane **302**, and the two-dimensional plane **302** is at infinity. However, since when the disparity of the stereoscopic display **120** is 0, when the equivalent is a two-dimensional plane, the two-dimensional plane is not at infinity, but on a display surface **122** of the display, and a distance is d, so in a situation of the same disparity amount, when images at other positions or at different depths are projected on the stereoscopic display **120**, the distance also changes, like as changing from infinity to d when the disparity is zero. Therefore, the disparity correction amount  $\delta_B$  needs to be added. That is to say, the image distance of the stereoscopic camera **200** at zero disparity is infinite, and the image distance of the stereoscopic display **120** at zero disparity is a finite value. Step **S120** includes calculating the disparity correction amount  $\delta_B$  according to the finite value.

$$m_m = \frac{\text{monitor\_pixels}}{\text{mm}} \quad (\text{Formula 11})$$

$$\frac{Z_m}{IPD_1} = \frac{d - Z_m}{\frac{1}{m_m}(u_{1L} - u_{1R} + \delta_B)} \quad (\text{Formula 12})$$

$$Z_m = \frac{d}{\frac{1}{m_m}(u_{1L} - u_{1R} + \delta_B) \cdot IPD_1 + 1} \quad (\text{Formula 13})$$

$$Z_m = Z_1 \quad (\text{Formula 14})$$

$$\frac{d}{\frac{1}{m_m}(u_{1L} - u_{1R} + \delta_B) \cdot IPD_1 + 1} = \frac{IPD_1}{\tan \frac{(u_{1L} - u_{1R})}{2PPD_{cm}}} \quad (\text{Formula 15})$$

$$\delta_B = m_m d \tan \frac{(u_{1L} - u_{1R})}{2PPD_{cm}} - m_m IPD_1 - (u_{1L} - u_{1R}) \quad (\text{Formula 16})$$

$$\text{Disparity}_m = \frac{(u_{mL} - u_{mR})}{m_m} = (u_{1L} - u_{1R} + \delta_B) \quad (\text{Formula 17})$$

As in FIG. 8,  $\delta_B$  is added to a disparity content projected on the stereoscopic display **120**. In addition, the pixel density of the stereoscopic display **120** is  $m_m$ , and monitor\_pixels is the number of pixels per mm of the stereoscopic display **120**. Assuming that the observer is the same, that is, the first observer **60** sees the stereoscopic display **120**, and a resulting stereoscopic depth is  $Z_m$ . From Formula 14, of which the objective is that a seen depth is equal to the depth when seeing the virtual reality display **300**, and then the difference amount  $\delta_B$  is obtained.

Formula 12 may be obtained from similar triangles, and  $Z_m$  may be obtained after arrangement, as Formula 13. Formula 15 may be obtained from Formula 14, and  $\delta_B$  may be obtained after arrangement, as Formula 16. A new disparity amount  $\text{Disparity}_m$  (i.e.,  $(u_{mL} - u_{mR})$ ) is as described in Formula 17.

FIG. 9 is a graph for calculating the disparity correction amount  $\delta_C$  in FIG. 2. Please refer to FIG. 3A and FIG. 9. When calculating the disparity correction amount  $\delta_C$ , assuming that it is the second observer **70** who sees the stereoscopic display **120**, and since the second observer **70** and the first observer **60** have different IPDs, which are  $IPD_2$  and  $IPD_1$  respectively, the correction amount  $\delta_C$  needs to be added.

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$$\frac{Z_2}{IPD_2} = \frac{d - Z_2}{\frac{1}{m_m}(u_{mL} - u_{mR} + \delta_C)} \quad (\text{Formula 18})$$

$$Z_m = \frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR} + \delta_C) \cdot IPD_2 + 1} \quad (\text{Formula 19})$$

$$Z_2 = Z_m \quad (\text{Formula 20})$$

$$\frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR} + \delta_C) \cdot IPD_2 + 1} = \frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR}) \cdot IPD_1 + 1} \quad (\text{Formula 21})$$

$$\delta_C = \left( \frac{IPD_2}{IPD_1} - 1 \right) (u_{mL} - u_{mR}) \quad (\text{Formula 22})$$

$$\text{Disparity}_2 = \frac{(u_{2L} - u_{2R})}{m_m} = (u_{mL} - u_{mR} + \delta_C) \quad (\text{Formula 23})$$

As in FIG. 12, a  $\delta_C$  is added and the IPD of the second observer **70** is  $IPD_2$ , and an stereoscopic visual distance is  $Z_2$ . Formula 18 may be obtained from similar triangles, and  $Z_2$  may be obtained after arrangement, as Formula 19. Since the object is for the second observer **70** and the first observer **60** to feel the same stereoscopic depth, as in Formula 20, Formula 21 may be obtained, and Formula 22 may be obtained after arrangement. The increased amount of the disparity is  $\delta_C$ . A new disparity  $\text{Disparity}_2$  (i.e.  $(u_{2L} - u_{2R})$ ) is as described in Formula 23.

After a series of derivations,  $\delta_A$ ,  $\delta_B$ , and  $\delta_C$  may be obtained. Then, the disparity of the second observer **70** may be obtained by adding  $\delta_A$ ,  $\delta_B$ , and  $\delta_C$  to the disparity photographed by the first observer **60**, and the stereoscopic depth seen by the two observers may be the same.

In an embodiment, the controller **110** is, for example, a central processing unit (CPU), a microprocessor, a digital signal processor (DSP), a programmable controller, a programmable logic device (PLD) or other similar devices or a combination of the devices, and the disclosure is not limited thereto. In addition, in an embodiment, each function of the controller **110** may be implemented as a plurality of program codes. These program codes are stored in a memory and are executed by the controller **110**. Alternatively, in an embodiment, each function of the controller **110** may be implemented as one or more circuits. The disclosure does not limit the implementation of the functions of the controller **110** by means of software or hardware.

In this embodiment, the device for adjusting a depth of a stereoscopic image **100** may be a notebook computer, and the controller **110** is, for example, the CPU of the notebook computer, and the stereoscopic display **120** is, for example, the display of the notebook computer, but the disclosure is not limited thereto. In other embodiments, the controller **110** may also be separated from the stereoscopic display **120** and be positioned on another server or desktop computer. In addition, there may be a plurality of controllers **110**, and a controller performing Step **S110** may be positioned on the stereoscopic camera **200**, while a controller performing Steps **S120** to **S140** may be positioned on a notebook computer, a server or a desktop computer.

FIG. 10 is a flowchart of a method for adjusting a depth of a stereoscopic image according to another embodiment of the disclosure, and FIG. 11 is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 10. Please refer to FIG. 10 and FIG. 11. The method for adjusting a depth of a stereoscopic

image of this embodiment is similar to the method for adjusting a depth of a stereoscopic image in FIG. 2, while a difference is as the following. In this embodiment, a distance B' between centers of the two lenses 210 of the stereoscopic camera 200 is adjustable, for example, may be adjusted by a mechanism. In Step S110' of the method for adjusting a depth of a stereoscopic image of this embodiment, the distance B' between centers of the two lenses 210 of the stereoscopic camera 200 is adjusted to be the same as the IPD  $IPD_1$  of the first observer 60, so that a stereoscopic image distance of an object photographed by the stereoscopic camera 200 is equal to the distance D1 of the first observer 60 seeing the object. In this way, after such adjustment, it may be considered that the disparity correction amount  $\delta_A$  is zero. In this case, in Step S140', there is no need to consider the disparity correction amount  $\delta_A$ , so step S140' may be to transmit a stereoscopic image of the object 80 photographed by the stereoscopic camera 200 which has been corrected by the disparity correction amount  $\delta_B$  and the disparity correction amount  $\delta_C$  to the stereoscopic display 120. In addition, the device for adjusting a depth of a stereoscopic image 100 of the embodiment of the disclosure may also be used to implement the method for adjusting a depth of a stereoscopic image shown in FIG. 10.

In summary, in the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the disparity correction amount  $\delta_A$  is obtained according to the distance between centers of the two lenses of the stereoscopic camera and the IPD of the first observer, the disparity correction amount  $\delta_B$  is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the disparity correction amount  $\delta_C$  is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the disparity correction amount  $\delta_A$ , the disparity correction amount  $\delta_B$ , and the disparity correction amount  $\delta_C$  is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect. In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the distance between centers of the two lenses of the stereoscopic camera is adjusted to be the same as the IPD of the first observer, the disparity correction amount  $\delta_B$  is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, a disparity correction amount  $\delta_C$  is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the disparity correction amount  $\delta_B$  and the disparity correction amount  $\delta_C$  is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect.

What is claimed is:

1. A method for adjusting a depth of a stereoscopic image, comprising:

obtaining a first disparity correction amount according to a distance between centers of two lenses of a stereoscopic camera and an interpupillary distance (IPD) of a first observer, wherein a stereoscopic image distance

formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object;

obtaining a second disparity correction amount according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera;

obtaining a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and

transmitting a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

2. The method for adjusting the depth of the stereoscopic image according to claim 1, further comprising calculating the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount by a pixel difference.

3. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera, wherein the method for adjusting the depth of the stereoscopic image further comprises displaying, by the virtual reality display, a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, so that a stereoscopic image distance of the stereoscopic image seen by the first observer through the virtual reality display is equal to a distance of the first observer directly seeing the object.

4. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

5. The method for adjusting the depth of the stereoscopic image according to claim 4, wherein obtaining the second disparity correction amount according to the difference between the image distances of the stereoscopic display and the stereoscopic camera at zero disparity comprises calculating the second disparity correction amount according to the finite value.

6. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the method for obtaining the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is performed through calculation with triangulation.

7. A device for adjusting a depth of a stereoscopic image, comprising:

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at least one controller configured to receive a stereoscopic image signal from a stereoscopic camera and to control a display content of a stereoscopic display, wherein the controller is configured to:

obtain a first disparity correction amount according to a distance between centers of two lenses of the stereoscopic camera and an IPD of a first observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object;

obtain a second disparity correction amount according to a difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera;

obtain a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and

transmit a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

8. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the controller is further configured to: calculate the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount by a pixel difference.

9. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera, wherein the controller is further configured to: display, by the virtual reality display, a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, so that a stereoscopic image distance of the stereoscopic image seen by the first observer through the virtual reality display is equal to a distance of the first observer directly seeing the object.

10. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

11. The device for adjusting the depth of the stereoscopic image according to claim 10, wherein obtaining the second disparity correction amount according to the difference between the image distances of the stereoscopic display and

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the stereoscopic camera at zero disparity comprises calculating the second disparity correction amount according to the finite value.

12. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the method for obtaining the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is performed through calculation with triangulation.

13. A method for adjusting a depth of a stereoscopic image, comprising:

adjusting a distance between centers of two lenses of a stereoscopic camera to be the same as an IPD of a first observer, so that a stereoscopic image distance of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object;

obtaining a first disparity correction amount according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera;

obtaining a second disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount and the second disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and

transmitting a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount to the stereoscopic display.

14. The method for adjusting the depth of the stereoscopic image according to claim 13, further comprising calculating the first disparity correction amount and the second disparity correction amount by a pixel difference.

15. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera.

16. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

17. The method for adjusting the depth of the stereoscopic image according to claim 16, wherein obtaining the first disparity correction amount according to the difference between the image distances of the stereoscopic display and the stereoscopic camera at zero disparity comprises calculating the first disparity correction amount according to the finite value.

18. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the method for obtaining the first disparity correction amount and the second disparity correction amount is performed through calculation with triangulation.

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